

Banking Transaction Monitoring System

Portfolio Project Summary

Objective: analyze transaction activity, monitor channel and monthly trends, review transaction statuses, and identify records that meet illustrative risk rules.

Technology stack: MySQL, SQL, Python (Pandas and Matplotlib), Power BI, DAX.

Dataset: 500 synthetic customers and 10,000 synthetic transactions. No real banking or personal data is used.

Metric	Value
Customers	500
Transactions	10,000
Successful transactions	9,315
Rule-flagged transactions	171
Total transaction value (all statuses)	37,229,265.82

Analytical questions

- How does successful transaction value change by month?
- Which transaction channels contribute the most value?
- What share of transactions fail or are reversed?
- Which transactions meet the sample rule-based risk criteria?
- How do transaction values differ by city and merchant category?

Important limitation

The risk_flag column is generated from simple amount/channel rules for learning purposes. It is not a trained or validated fraud model and must not be represented as one.